Project 2 Report: Employee Attrition

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Abstract This project develops a blank.

Introduction

The objective of this project is to develop a predictive model capable of determining which employees are most likely to leave. This can allow the business to focus their retention efforts on high-risk employees.

The business decision addressed by this analysis is:

> How much financial value does using this predictive model create compared with not using one?

The resulting model can be used to identify employees that appear likely to leave.

Business Objective

The primary objective is to accurately determine which employees are likely to quit before they do.

A practical utility function for this problem is:

$$U = \begin{cases} 0.60 \times 40000 - 8000 = 16000 & \text{if a retention intervention is provided and prevents attrition} \\ -8000 & \text{if a retention intervention is provided unnecessarily} \\ 0 & \text{if no intervention is provided and the employee leaves} \\ 0 & \text{if no intervention is provided and the employee stays} \end{cases}$$

The utility function reflects the incremental financial impact of acting on the model's recommendations. A true positive represents the expected net savings from successfully retaining an employee, while a false positive represents the cost of an unnecessary retention intervention. False negatives and true negatives are assigned zero utility because they represent the baseline outcome in which no intervention occurs.

assumptions:

- Cost of employee turnover: $40,000
- Cost of retention intervention: $8,000
- Retention intervention success rate: 60%

Plan

The analysis follows the regression modeling workflow:

1. Define the business problem.
2. Acquire and clean employee data.
3. Explore relationships between candidate predictors and attrition.
4. Build an interpretable regression model.
5. Evaluate assumptions and model fit.
6. Generate predictions and business recommendations.

The ideal dataset would contain:

- Age
- Attrition
- Salary
- Years at the company
- Overtime
- Job role
- Job satisfication score

Due to the data being fake generated data, it was more than ideal. It included everything we thought of and much more.

Build

Data Source

The dataset was obtained from Kaggle at https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset and contains a fictional dataset created by IBM data scientists.

Variables initially available included 35 different point including notably:

- Age
- Attrition
- Department
- Gender
- Job Role
- Job Satisfaction
- Martial Status
- Monthly Income
- Number of Companies Worked
- Overtime

- Years Since Last Promotion
- Year at Company
- Years in Current Role

Data Cleaning

The following preprocessing steps were performed:

- Converted Attrition to a binary outcome where Yes = 1

```
import os
import numpy as np
import polars as pl
import pandas as pd
import seaborn.objects as so
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf

from patsy import dmatrices
from sklearn.base import clone
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import (
  confusion_matrix,
  accuracy_score,
  precision_score,
  recall_score,
  roc_auc_score
)
from statsmodels.stats.outliers_influence import variance_inflation_factor
# from sklearn.utils import resample

pl.Config.set_tbl_rows(-1)
pl.Config.set_tbl_cols(-1)
pl.Config.set_fmt_str_lengths(100)

employee_data = (pl.read_csv(os.path.join('data', 'HR-Employee-Attrition.csv'))
  # Make the outcome binary where 'Yes' = 1
  .with_columns(
    pl.when(pl.col('Attrition') == 'Yes').then(1)
    .when(pl.col('Attrition') == 'No').then(0)
```

```
    .alias('Left')
  )
  .select(pl.exclude(['Attrition', 'Department', 'EmployeeNumber', 'DailyRate',
'MonthlyIncome', 'EmployeeCount', 'Over18', 'PerformanceRating']))
)

# Print unique values and their counts
# for col in employee_data.columns:
#   print(f"\n{'='*60}")
#   print(f"Column: {col}")
#   print(f"{'='*60}")

#   print(
#     employee_data
#     .group_by(col)
#     .len()
#     .sort("len", descending=True)
#   )

predictors = [
  "Age",
  "C(BusinessTravel)",
  # "C(Department)", # job role perfectly maps to certain departments except
manager
  "DistanceFromHome",
  "C(EducationField)", # This seems to be redundant.
  "Education",
  "EnvironmentSatisfaction",
  "C(Gender)",
  "HourlyRate", # Hourly rate seems to group employees together better than other
similar rate columns that tend to only match one or few employees.
  "JobInvolvement",
  "JobLevel",
  "C(JobRole)",
  "JobSatisfaction",
  "C(MaritalStatus)",
  "MonthlyRate", # may be more linear and likely redundant with hourly rate
  "NumCompaniesWorked",
  "C(OverTime)",
  "PercentSalaryHike",
  "RelationshipSatisfaction",
  "TotalWorkingYears", # may correlate strongly with Age
  "TrainingTimesLastYear",
```

```
  "WorkLifeBalance",
  "YearsAtCompany",
  "YearsInCurrentRole",
  "YearsSinceLastPromotion",
  "YearsWithCurrManager"
]
```

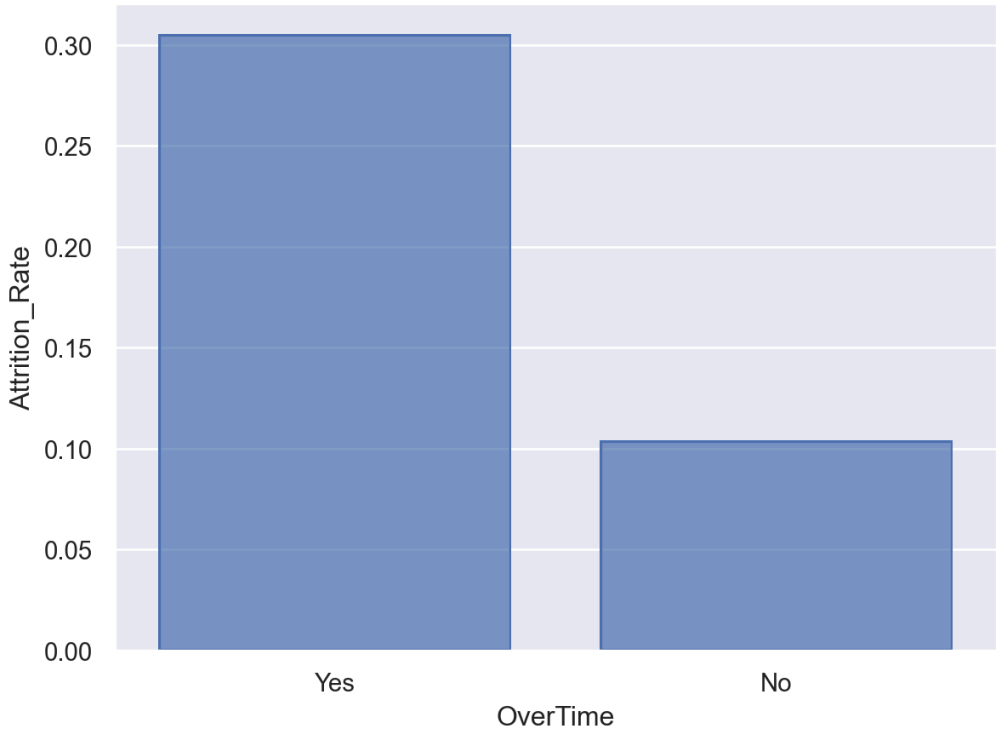
Explore

Exploratory Data Analysis

Initial visualizations focused on the relationships between attrition and major numerical predictors.

Overtime vs Attrition

```
overtime_rate = (
    employee_data
    .group_by("OverTime")
    .agg(Attrition_Rate=pl.col("Left").mean())
)

so.Plot(overtime_rate.to_pandas(),
        x="OverTime",
        y="Attrition_Rate") \
    .add(so.Bar())
```

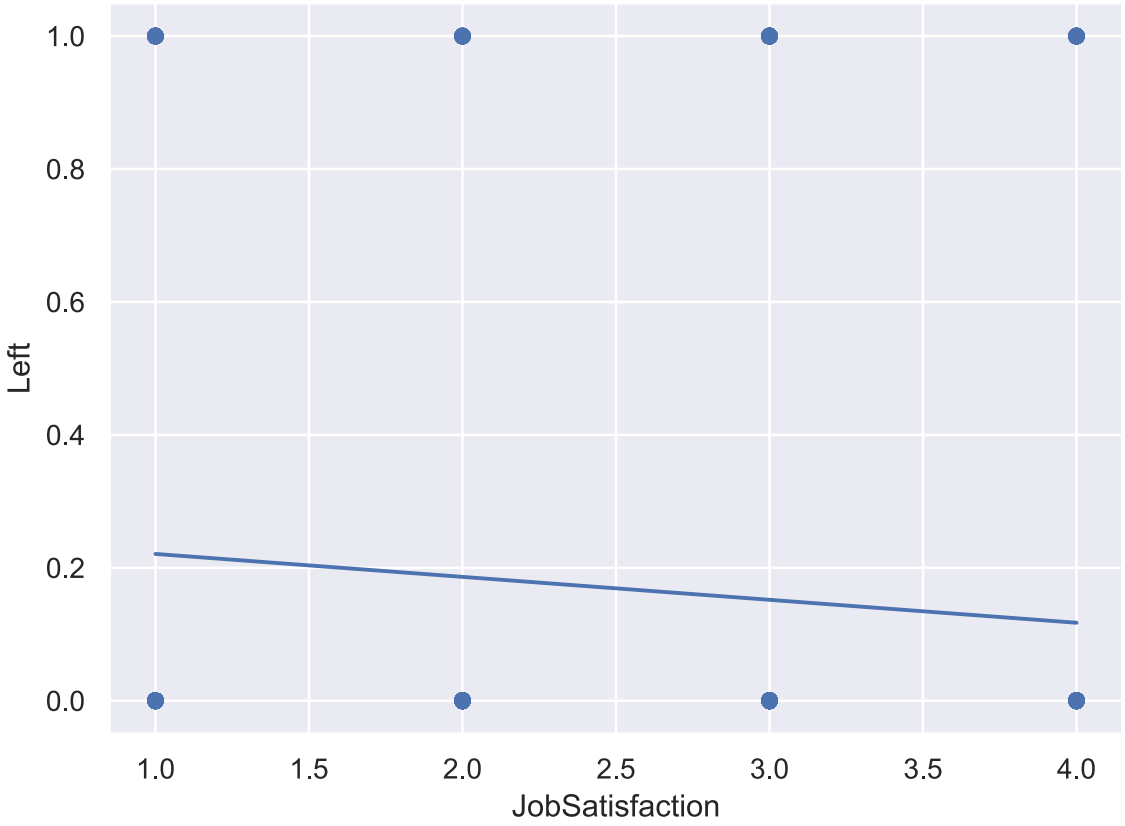


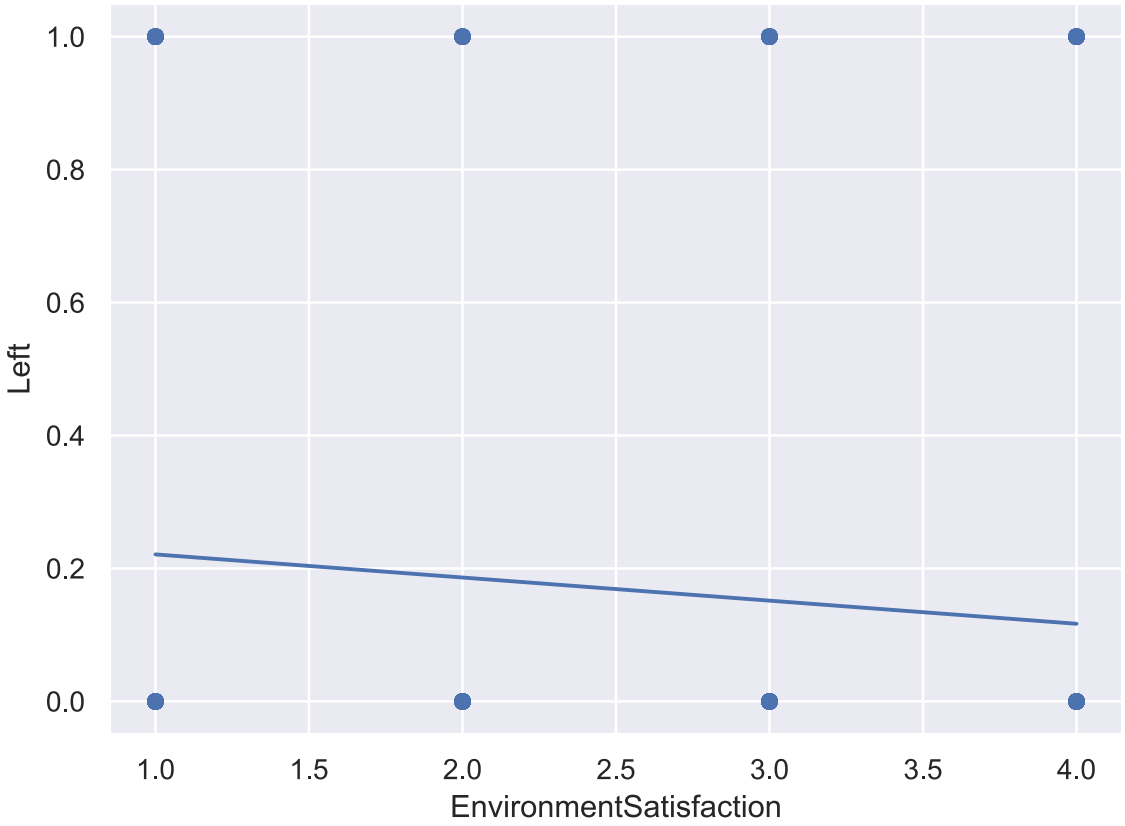
This looks like employees that work overtime are more likely to leave.

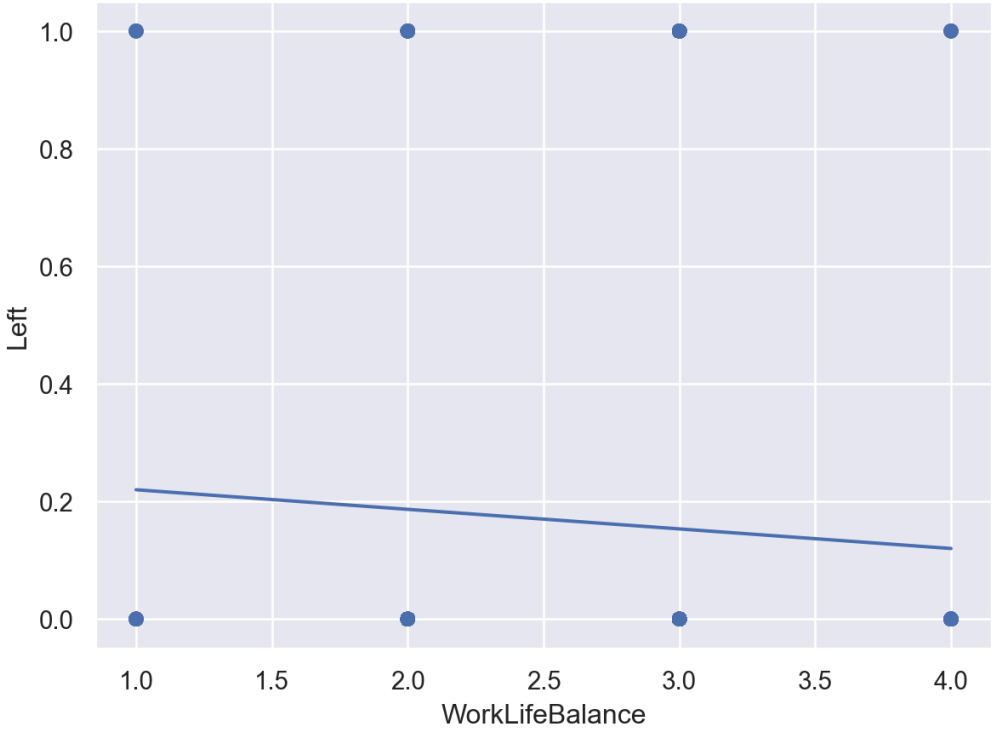
Satisfaction vs Attrition

```
(
    so.Plot(employee_data.to_pandas(),
            x="JobSatisfaction",
            y="Left")
    .add(so.Dot(alpha=.20))
    .add(so.Line(), so.PolyFit(order=1))
).show()

(
    so.Plot(employee_data.to_pandas(),
            x="EnvironmentSatisfaction",
            y="Left")
    .add(so.Dot(alpha=.20))
    .add(so.Line(), so.PolyFit(order=1))
).show()

(
    so.Plot(employee_data.to_pandas(),
            x="WorkLifeBalance",
```

```
            y="Left")
    .add(so.Dot(alpha=.20))
    .add(so.Line(), so.PolyFit(order=1))
)
```



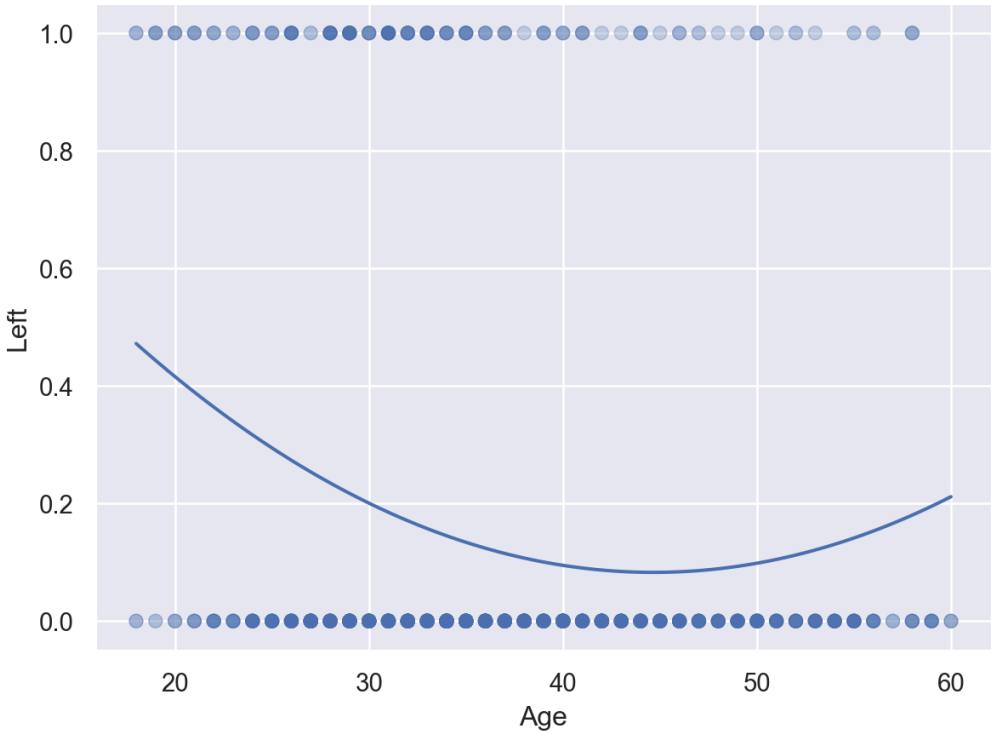

1.0
0.8
0.6
0.4
0.2
0.0
Left
1.0
1.5
2.0
2.5
3.0
3.5
4.0
EnvironmentSatisfaction



These graphs show what we would expect, that as employees report being more satisfied or having better work/life balance, they are less likely to leave.

Age vs Attrition

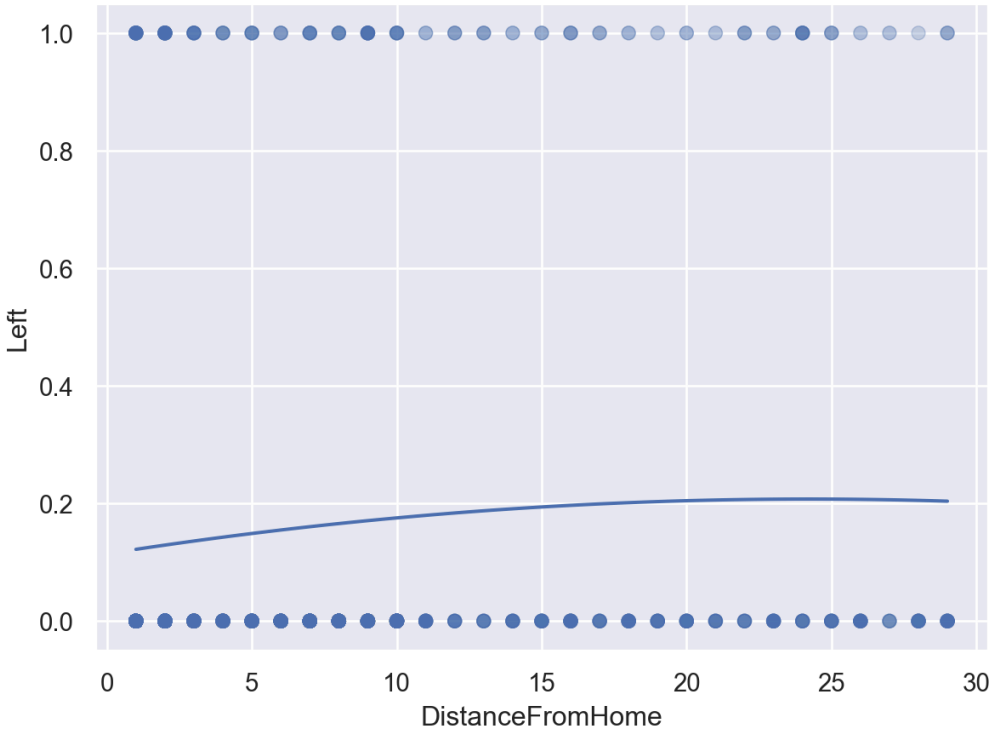
```
(
    so.Plot(employee_data.to_pandas(),
            x="Age",
            y="Left")
    .add(so.Dot(alpha=.15))
    .add(so.Line(), so.PolyFit(order=2))
)
```



We can see that younger employees are generally more likely to leave, but this starts to reverse around age 45. Retirement is likely a factor, but also employees are likely changing jobs for more senior roles.

Commute vs Attrition

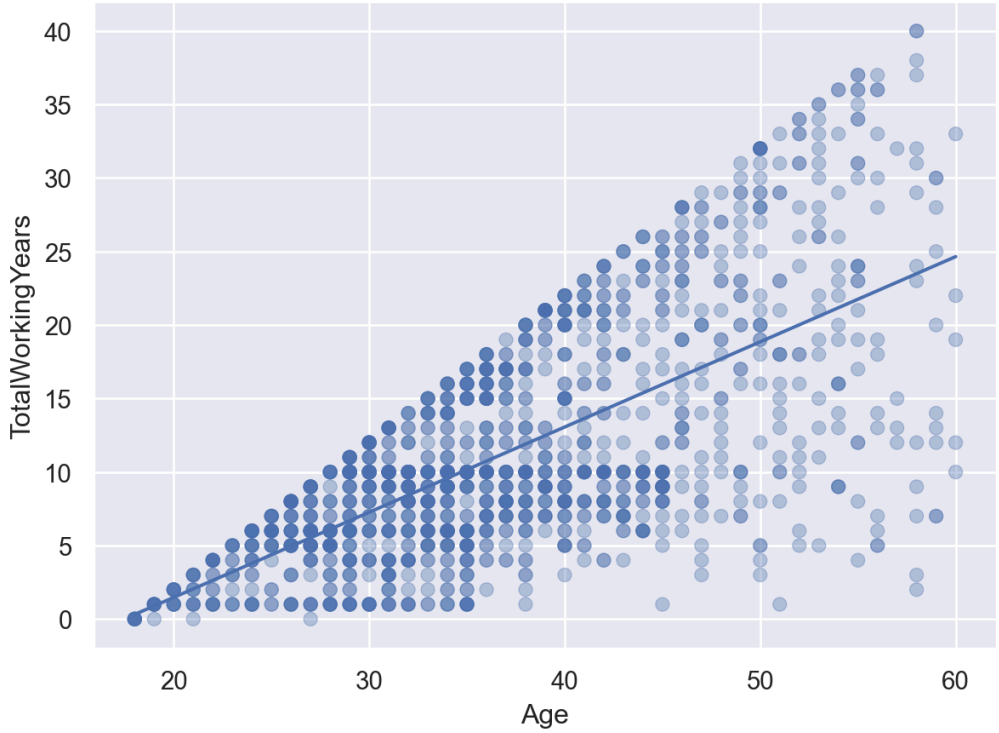
```
(
    so.Plot(employee_data.to_pandas(),
            x="DistanceFromHome",
            y="Left")
    .add(so.Dot(alpha=.15))
    .add(so.Line(), so.PolyFit(order=2))
)
```



There is a fairly weak correlation between how far someone lives from work and if they are likely to leave. It is interesting that the trend appears to level off so it seems like it's only a factor within a certain range.

Age vs Total Working Years

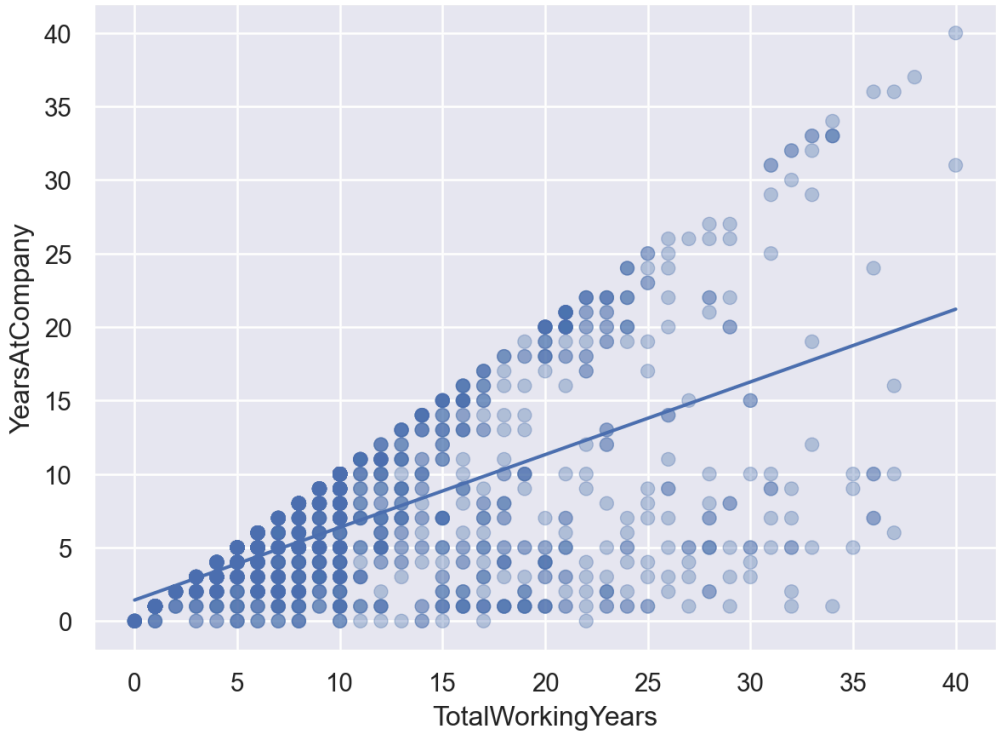
```
(
    so.Plot(employee_data.to_pandas(),
            x="Age",
            y="TotalWorkingYears")
    .add(so.Dot(alpha=.35))
    .add(so.Line(), so.PolyFit(order=1))
)
```



We thought there would be a strong relationship in the data that could cause issues between employee age and total working years. It looks like there is a relationship, but the correlation is not nearly as strong as we thought.

Years at Company vs Total Working Years

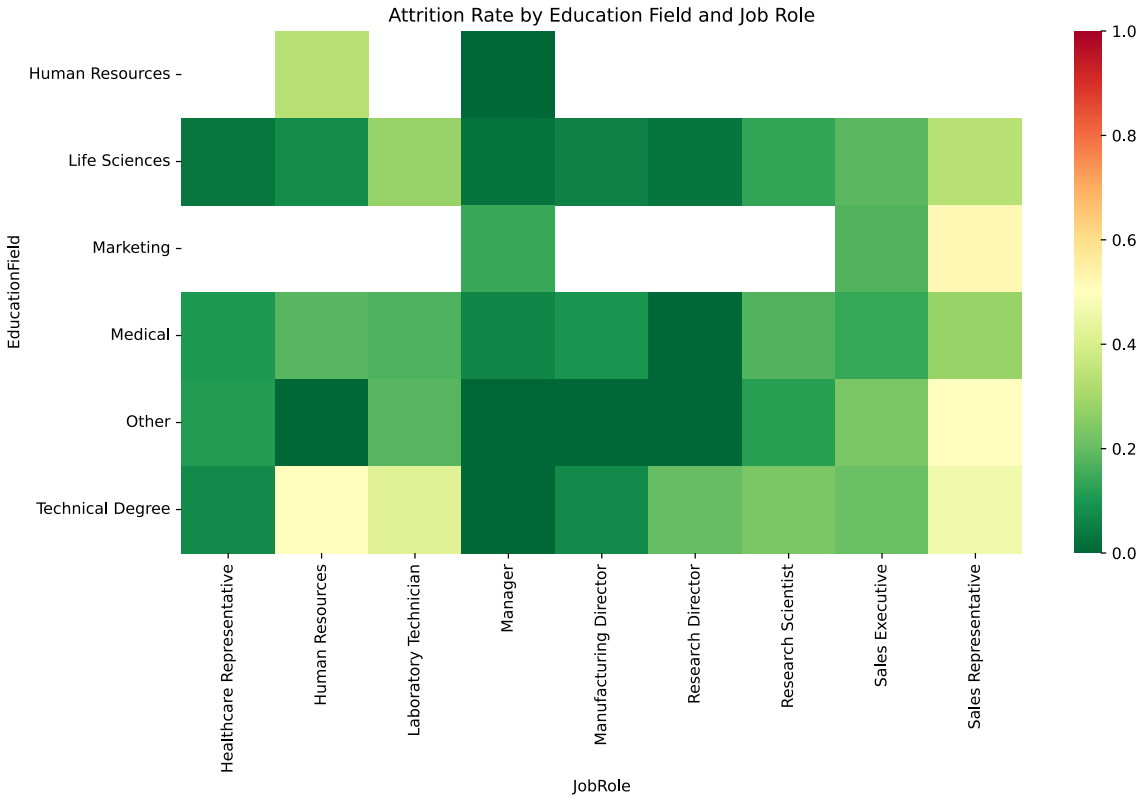
```
(
    so.Plot(employee_data.to_pandas(),
            x="TotalWorkingYears",
            y="YearsAtCompany")
    .add(so.Dot(alpha=.35))
    .add(so.Line(), so.PolyFit(order=1))
)
```



It looks like there is a relationship between years at the company and total working years, but not a very strong one.

Education Field vs Job Role

```
rate = (
    employee_data
    .group_by(["EducationField", "JobRole"])
    .agg(attrition_rate=pl.col("Left").mean())
    .to_pandas()
)
pivot_rate = rate.pivot(
    index="EducationField",
    columns="JobRole",
    values="attrition_rate"
)
plt.figure(figsize=(12,6))
sns.heatmap(pivot_rate, cmap="RdYlGn_r", vmin=0, vmax=1)
plt.title("Attrition Rate by Education Field and Job Role")
plt.show()
```

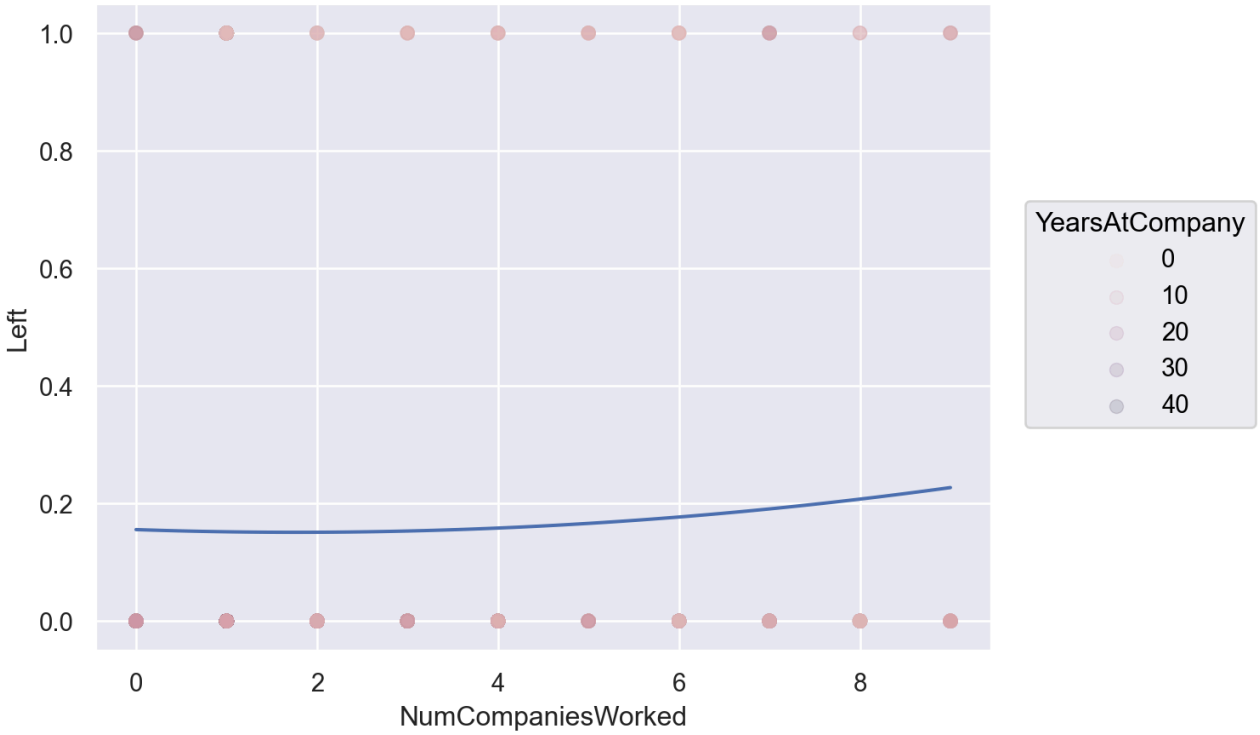


The heatmap reveals a strong structural relationship between education field and job role, indicating that employees are not randomly distributed across roles. Instead, specific education backgrounds concentrate within particular job families. This dependency explains multicollinearity we later observed in the initial logistic regression model.

Attrition varies more meaningfully across job roles than across education fields, suggesting that occupational role is a stronger predictor of turnover than educational background.

Number of Companies and Attrition

```
(
    so.Plot(employee_data.to_pandas(),
            x="NumCompaniesWorked",
            y="Left")
    .add(so.Dot(alpha=.15), color="YearsAtCompany")
    .add(so.Line(), so.PolyFit(order=2))
)
```



This is an interesting graph because it shows that employees that have worked are more companies are more likely to leave. This could be explained by the fact that employees that have worked at more companies are more likely to be looking for new opportunities. It also shows that employees that have been at the company for a long time are less likely to leave, which makes sense.

Key Findings

- Found this.
- Found that.

Reconcile

Sampling Strategy

A stratified sampling approach was used to evenly split test and training data with even attrition and non-attrition employees.

Assumptions

Model Refinements

Several iterations of the model were evaluated.

Adjustments included:

- abcd
- GDs.

- Evaluating Variance Inflation Factors (VIFs).

Hyper-parameter Tuning

```
employee_data = employee_data.to_dummies(
  columns=[
    'BusinessTravel',
    'EducationField',
    'Gender',
    'JobRole',
    'MaritalStatus',
    'OverTime'
  ]
).to_pandas()

# This is a utility function meant to maximize the use of money to retain employees
# It is based on our assumptions:
# * Cost of employee turnover: $40,000
# * Cost of retention intervention: $8,000
# * Retention intervention success rate: 60%
def utility(matrix):
  true_negative, false_positive, false_negative, true_positive = matrix.ravel()

  return (
    true_positive * 16000 +
    false_positive * (-8000) +
    false_negative * 0 +
    true_negative * 0
  )

X = employee_data.drop(columns="Left")
y = employee_data["Left"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Candidate cutoff probabilities and c values
c_values = np.logspace(-2, 2, 15)
thresholds = np.arange(0.20, 0.61, 0.05)
```

```
models = {
  "Ridge": LogisticRegression(l1_ratio=0),
  "Lasso": LogisticRegression(
    l1_ratio=1,
    solver="saga",
    max_iter=5000,
    random_state=42
  ),
  "Elastic": LogisticRegression(
    solver="saga",
    l1_ratio=0.5,
    max_iter=5000,
  )
}

# Setup proper 5-fold cross-validation object
kFolds = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

results = []

# We will try all 3 different models to see which one seems the best
for model_name, base_model in models.items():

  # We are going to tune our cutoff probability and c value at the same time
  best = {
    "C": None,
    "Threshold": None,
    "Utility": -np.inf,
    "Accuracy": 0,
    "Precision": 0,
    "Recall": 0,
    "AUC": 0
  }

  # Loop through each regularization parameter (C)
  for C in c_values:
    for threshold in thresholds:
      fold_utilities = []
      fold_accuracy = []
      fold_precision = []
      fold_recall = []
      fold_auc = []
```

```
# Iterate through the 5 distinct validation folds
for train_idx, val_idx in kFolds.split(X_train, y_train):

  # Split using the index arrays generated by StratifiedKFold
  X_train_fold = X_train.iloc[train_idx]
  X_val_fold = X_train.iloc[val_idx]

  y_train_fold = y_train.iloc[train_idx]
  y_val_fold = y_train.iloc[val_idx]

  # grab our frequentist logistic regression model and set C
  model = Pipeline([
    ("scaler", StandardScaler()),
    ("logistic", clone(base_model).set_params(C=C))
  ])

  model.fit(X_train_fold, y_train_fold)

  probabilities = model.predict_proba(X_val_fold)[:,1]

  predictions = (probabilities >= threshold).astype(int)

  cm = confusion_matrix(y_val_fold, predictions)

  fold_utilities.append(utility(cm))
  fold_accuracy.append(accuracy_score(y_val_fold, predictions))
  fold_precision.append(
      precision_score(y_val_fold, predictions, zero_division=0)
  )
  fold_recall.append(
      recall_score(y_val_fold, predictions, zero_division=0)
  )

  # AUC should use probabilities, not binary predictions
  fold_auc.append(
      roc_auc_score(y_val_fold, probabilities)
  )

avg_utility = np.mean(fold_utilities)
avg_accuracy = np.mean(fold_accuracy)
avg_precision = np.mean(fold_precision)
avg_recall = np.mean(fold_recall)
```

```
      avg_auc = np.mean(fold_auc)

      if avg_utility > best["Utility"]:
        best["C"] = C
        best["Threshold"] = threshold
        best["Utility"] = avg_utility
        best["Accuracy"] = avg_accuracy
        best["Precision"] = avg_precision
        best["Recall"] = avg_recall
        best["AUC"] = avg_auc

      # It's very possible we tie, if so go by accuracy
      elif (
          avg_utility == best["Utility"]
          and avg_accuracy > best["Accuracy"]
      ):
        best["C"] = C
        best["Threshold"] = threshold
        best["Utility"] = avg_utility
        best["Accuracy"] = avg_accuracy
        best["Precision"] = avg_precision
        best["Recall"] = avg_recall
        best["AUC"] = avg_auc

  # Save the best settings for this model
  results.append({
    "Model": model_name,
    **best
  })

# Identify the best cutoff probability, best C, and best model
results = pd.DataFrame(results)

results.sort_values(
    "Utility",
    ascending=False
).head()
```

	Model	C	Threshold	Utility	Accuracy	Precision	Recall	AUC
0	Ridge	0.071969	0.35	260800.0	0.885204	0.671458	0.568421	0.842234
2	Elastic	0.517947	0.40	260800.0	0.889452	0.701604	0.542105	0.841832
1	Lasso	1.000000	0.40	257600.0	0.887753	0.690919	0.542105	0.841591

Elastic Net achieved the highest cross-validated business utility (tied with Ridge), the highest overall accuracy, and the highest precision while maintaining an AUC comparable to the other models. Although Ridge produced slightly higher recall, the difference was small. Therefore, Elastic Net was selected as the final model because it provided the best overall balance of predictive performance and business value.

The similarity in performance suggests that the employee attrition dataset contains a relatively stable set of predictors with limited multicollinearity. Consequently, the choice of regularization method had only a modest impact on predictive performance. Our removal from the data of some predictors where we saw or predicted multicollinearity may have also contributed.

```
final_c = 0.517947
final_probability = 0.40

final_model = Pipeline([
    ("scaler", StandardScaler()),
    ("classification", LogisticRegression(
        solver="saga",
        l1_ratio=0.5,
        C=final_c,
        max_iter=5000,
        random_state=42
    ))
])

final_model.fit(X_train, y_train)

p_test = final_model.predict_proba(X_test)[:, 1]
y_pred = (p_test >= final_probability).astype(int)
final_accuracy = accuracy_score(y_test, y_pred)

final_accuracy

cm = confusion_matrix(y_test, y_pred)
print(utility(cm))

# formula = "Left ~ " + " + ".join(predictors)

# fit_01 = smf.logit(
#     formula,
#     data=X_train
# ).fit()

# print(fit_01.summary())
```

```
# # Create the design matrix used by the model
# y, X = dmatrices(formula, data=pb_train, return_type="dataframe")

# # Calculate VIF
# vif = pd.DataFrame({
#     "Variable": X.columns,
#    "VIF": [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]
# })

# print(vif.sort_values("VIF", ascending=False))
```

```
232000
```

Given the high P values, we should consider removing these from our model

Variable	p-value
Education 2–5	0.60–0.94
HourlyRate	0.656
PercentSalaryHike	0.584
YearsAtCompany	0.704

Then these next

Variable	p-value
JobLevel 3	0.936
JobLevel 4	0.528
JobLevel 5	0.466
JobRole Manager	0.846
JobRole Research Scientist	0.834
Manufacturing Director	0.441

Fit

Final Model

The final model included:

- Param 1
- Param 2

```
# fit_log.summary()
```

The fitted model produced:

- $R^2 = 0.818$
- Adjusted $R^2 = 0.812$
- F-statistic = 141.6
- $p < 0.001$

indicating strong overall explanatory power.

Important Continuous Predictors

Variable	Coefficient
OverTime	2.07
Business Travel	2.02/1.05
Marital Status	1.60

OverTime
Employees working overtime have roughly 8 times the odds of attrition compared with employees who do not, holding the other variables constant:

$$e^{2.07} \approx 7.95$$

Business Travel
Employees traveling frequently have roughly 7.5 times the odds of attrition compared to employees who do not travel, holding the other variables constant:

$$e^{2.02} \approx 7.5$$

Employees traveling rarely also have roughly 2.9 times the odds of attrition compared to employees who do not travel, holding the other variables constant:

$$e^{1.05} \approx 2.9$$

Marital Status
Single employees have roughly 5 times the odds of attrition compared to divorced employees:

$$e^{1.60} \approx 5$$

Evaluate

Residual Analysis
Residual diagnostics indicated:

- No severe violations of linearity.
- Acceptable residual symmetry.

- Some heavy tails remain.
- No extreme multicollinearity concerns.

The largest VIF values were associated with popular truck categories but remained below commonly cited thresholds requiring corrective action.

Limitations

Several limitations should be noted:

- No MSRP information was available.
- Trim level information was not incorporated.
- Accident history was unavailable.
- Maintenance records were unavailable.
- Auction data may not represent all vehicle markets.

These limitations likely explain a portion of the unexplained variance.

Predict

Based off of the given results, we do see that the model does give more value to vehicles in better condition and those that are newer. I'm slightly surprised that it gives more value to certain models over others but it does make sense considering that certain models are more sought after than others. It does indicate that trucks retain more value over sedans. These predictions seem reasonable and can be used to inform purchasing decisions.

Business Implications

The model suggests:

1. Vehicle age is the strongest continuous driver of depreciation.
2. Mileage has a significant but smaller effect.
3. Vehicle model contributes substantial value differences.
4. Trucks generally retain value better than passenger sedans.

Recommendations

Organizations purchasing used vehicles should:

- Prioritize lower-age vehicles when possible.
- Consider mileage alongside age rather than in isolation.
- Use model-specific adjustments when evaluating value.
- Compare observed asking prices against model predictions to identify unusually favorable purchasing opportunities.

Future Work

Future improvements could include:

1. Incorporating original MSRP.
2. Including trim-level information.
3. Adding accident-history variables.
4. Incorporating local economic indicators.

Conclusion

The final regression model explains approximately 82% of the variation in used vehicle selling prices and provides a practical framework for vehicle valuation. Vehicle age, mileage, model, and geographic location all contribute significantly to price determination. The resulting model can support data-driven purchasing and pricing decisions while highlighting areas for future model enhancement.